Sulabham · APPSC AEE 2026 — Module 4 Test: Building Materials

Paper 3 Civil · original practice questions in the pattern of APPSC AEE Paper III Civil, 22 Aug 2023 · take at the end of Week 7

Time: 50 minutes (exam pace: 1 minute per question) **Marks:** +1 correct, −⅓ wrong, 0 unattempted
Calculator: not allowed
Write your answers on a separate sheet. Mark a question only if you can rule out at least two options.
Online version with timer: sulabham.in/AEE_Module4_Test.html

Questions

A. Timber: Structure, Moisture and Strength

1. Assertion (A): On drying below the fibre saturation point, timber shrinks very little along its length.
Reason (R): Loss of bound water makes the cell walls thinner, which reduces the width of the fibres far more than their length.

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) Both A and R are false

2. A piece of timber weighs 1.80 kg at a moisture content of 20 %. Its oven-dry weight is:

(a) 1.44 kg (b) 1.50 kg (c) 1.60 kg (d) 1.62 kg

3. Match List-I (part of an exogenous tree) with List-II (description):

List-I	List-II
A. Cambium	1. Innermost, first-formed central part
B. Pith	2. Outer light-coloured rings that carry sap
C. Medullary rays	3. Thin layer of growing tissue just inside the bark
D. Sapwood	4. Radial bands that hold the annual rings together

(a) A-1, B-3, C-4, D-2 (b) A-3, B-1, C-2, D-4 (c) A-3, B-1, C-4, D-2 (d) A-4, B-1, C-3, D-2

4. Study the statement and the conclusions and select the most appropriate option.
Statement: A timber plank is dried from its fibre saturation point to a moisture content of 12 %.
Conclusion 1: Its strength increases.
Conclusion 2: Its dimensions remain unchanged.

(a) Only conclusion 1 is correct
(b) Only conclusion 2 is correct
(c) Both conclusions 1 and 2 are correct
(d) Neither conclusion 1 nor 2 is correct

5. As per IS 883, structural timbers are classified into Groups A, B and C on the basis of their:

(a) density at 12 % moisture content
(b) compressive strength parallel to the grain
(c) natural durability
(d) modulus of elasticity

B. Timber: Defects, Seasoning, Preservation and Plywood

6. S1: Dry rot attacks timber kept in dry, well-ventilated places.
S2: Wet rot is caused by alternate wetting and drying of timber.

(a) Both S1 and S2 are true (b) S1 is true and S2 is false (c) S1 is false and S2 is true (d) Both S1 and S2 are false

7. Match List-I (seasoning defect) with List-II (description):

List-I	List-II
A. Bow	1. Curvature across the width of the board
B. Cup	2. Internal cracks not visible at the surface
C. Honeycombing	3. Surface dried and set while the interior is still wet
D. Case hardening	4. Curvature along the length of the board

(a) A-1, B-4, C-2, D-3 (b) A-4, B-1, C-3, D-2 (c) A-2, B-1, C-4, D-3 (d) A-4, B-1, C-2, D-3

8. ASCU preservative for timber consists of hydrated arsenic pentoxide, copper sulphate and potassium dichromate in the ratio:

(a) 1 : 2 : 3 (b) 1 : 3 : 4 (c) 3 : 4 : 1 (d) 4 : 3 : 1

9. Which of the following statements about plywood is INCORRECT?

(a) It is made of an odd number of veneers (b) It has nearly equal strength along its length and width

(c) It shrinks and swells much less than solid timber (d) The grain of each veneer is parallel to the grain of the next

10. Assertion (A): Air seasoning cannot normally reduce the moisture content of timber much below about 15 %.

Reason (R): Air seasoning is done in a closed chamber with controlled temperature and humidity.

(a) Both A and R are true and R is the correct explanation of A (b) Both A and R are true, but R is not the correct explanation of A

(c) A is true, but R is false (d) Both A and R are false

C. Bricks

11. The number of modular bricks (190 mm × 90 mm × 90 mm, with 10 mm mortar joints) required for a wall 4 m long, 3 m high and 200 mm thick is:

(a) 1000 (b) 1200 (c) 1440 (d) 2400

12. Match List-I (defect in bricks) with List-II (cause):

List-I	List-II
A. Chuffs	1. Soluble alkali salts in the clay
B. Efflorescence	2. Rain water falling on hot bricks
C. Black core	3. Nodules of lime slaking after burning
D. Lime blowing	4. Carbonaceous matter not burnt out

(a) A-2, B-1, C-4, D-3 (b) A-1, B-2, C-4, D-3 (c) A-2, B-1, C-3, D-4 (d) A-2, B-4, C-1, D-3

13. S1: Excess of alumina in brick earth makes raw bricks shrink and warp during drying.

S2: Excess of silica in brick earth makes the bricks brittle.

(a) Both S1 and S2 are true (b) S1 is true and S2 is false (c) S1 is false and S2 is true (d) Both S1 and S2 are false

14. A brick weighs 2.50 kg when dry, 2.85 kg after 24 hours in cold water and 2.95 kg after 5 hours in boiling water. Its saturation coefficient is about:

(a) 0.14 (b) 0.97 (c) 1.29 (d) 0.78

15. Study the statement and the conclusions and select the most appropriate option.

Statement: A lot of common burnt clay bricks is designated class 10 as per IS 1077.

Conclusion 1: The average compressive strength of the bricks is not less than 10 N/mm².

Conclusion 2: The water absorption of the bricks must not exceed 15 %.

(a) Only conclusion 1 is correct (b) Only conclusion 2 is correct

(c) Both conclusions 1 and 2 are correct (d) Neither conclusion 1 nor 2 is correct

D. Brick Masonry and Building Stones

16. Assertion (A): Mortar much stronger than the bricks is not recommended for brick masonry.

Reason (R): The strength of masonry increases in direct proportion to the strength of the mortar.

(a) Both A and R are true and R is the correct explanation of A (b) Both A and R are true, but R is not the correct explanation of A

(c) A is true, but R is false (d) Both A and R are false

17. A masonry wall has an effective height of 3.0 m and an effective thickness of 200 mm. Its slenderness ratio is:

(a) 6.7 (b) 15 (c) 27 (d) 60

18. Match List-I (stone) with List-II (geological and chemical class):

List-I	List-II
A. Granite	1. Metamorphic, calcareous
B. Sandstone	2. Igneous, siliceous
C. Slate	3. Sedimentary, siliceous
D. Marble	4. Metamorphic, argillaceous

(a) A-3, B-2, C-4, D-1 (b) A-2, B-3, C-1, D-4 (c) A-2, B-4, C-3, D-1 (d) A-2, B-3, C-4, D-1

19. In Brard's test for the frost resistance of a building stone, the specimen is repeatedly immersed in a solution of:

(a) dilute hydrochloric acid (b) sodium sulphate (c) sodium hydroxide (d) copper sulphate

20. Which of the following statements about brick masonry is INCORRECT?

(a) Thicker mortar joints reduce the strength of masonry
(b) Bricks are soaked before laying to prevent suction of water from the mortar
(c) Permissible stress reduces with increase in slenderness ratio
(d) In English bond, headers and stretchers alternate in the same course

E. Cement: Compounds, Hydration and Types

21. Study the statement and the conclusions and select the most appropriate option.

Statement: Cement P contains 55 % C_3S and 18 % C_2S; cement Q contains 40 % C_3S and 33 % C_2S. They are otherwise alike.

Conclusion 1: Cement P gains strength faster in the first week.

Conclusion 2: Cement P liberates less heat of hydration than cement Q.

(a) Only conclusion 1 is correct
(b) Only conclusion 2 is correct
(c) Both conclusions 1 and 2 are correct
(d) Neither conclusion 1 nor 2 is correct

22. Match List-I (cement) with List-II (most suitable use):

List-I	List-II
A. Rapid hardening cement	1. Mass concrete in dams
B. Low heat cement	2. Concreting in running water
C. Sulphate resisting cement	3. Urgent road repair work
D. Quick setting cement	4. Foundations in soils containing sulphates

(a) A-1, B-3, C-4, D-2 (b) A-3, B-1, C-2, D-4 (c) A-3, B-1, C-4, D-2 (d) A-2, B-1, C-4, D-3

23. The compound of Portland cement that reacts first with water and liberates the greatest heat per gram is:

(a) C_3S (b) C_2S (c) C_3A (d) C_4AF

24. S1: Portland pozzolana cement evolves less heat of hydration than ordinary Portland cement.

S2: Portland pozzolana cement gains strength faster than ordinary Portland cement in the first few days.

(a) Both S1 and S2 are true (b) S1 is true and S2 is false (c) S1 is false and S2 is true (d) Both S1 and S2 are false

25. Assertion (A): Concrete made with Portland pozzolana cement has less free calcium hydroxide than concrete made with ordinary Portland cement.

Reason (R): Pozzolana combines with calcium hydroxide in the presence of water to form additional calcium silicate hydrate.

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) Both A and R are false

F. Cement: Tests

26. The standard consistency of a cement is 28 %. The water to be added to a mix of 200 g of cement and 600 g of standard sand for making compressive-strength cubes is:

(a) 20 g (b) 56 g (c) 80 g (d) 60 g

27. Match List-I (test on ordinary Portland cement) with List-II (limit):

List-I	List-II
A. Initial setting time	1. Not more than 0.8 %
B. Final setting time	2. Not less than 30 minutes
C. Le Chatelier expansion	3. Not more than 10 mm
D. Autoclave expansion	4. Not more than 600 minutes

(a) A-2, B-4, C-1, D-3 (b) A-4, B-2, C-3, D-1 (c) A-2, B-3, C-4, D-1 (d) A-2, B-4, C-3, D-1

28. The minimum compressive strength of 43 grade ordinary Portland cement at 7 days is:

(a) 23 N/mm² (b) 27 N/mm² (c) 33 N/mm² (d) 37 N/mm²

29. Taking the bulk density of cement as 1440 kg/m³, the volume of 10 bags of cement is about:

(a) 0.144 m³ (b) 0.347 m³ (c) 0.500 m³ (d) 0.720 m³

30. Study the statement and the conclusions and select the most appropriate option.

Statement: A sample of cement shows an expansion of 14 mm in the Le Chatelier test.

Conclusion 1: The cement is not ground fine enough.

Conclusion 2: The cement contains excess free lime.

(a) Only conclusion 1 is correct (b) Only conclusion 2 is correct

(c) Both conclusions 1 and 2 are correct (d) Neither conclusion 1 nor 2 is correct

G. Mortar and Lime

31. Sand on a site has bulked by 25 %. The volume of dry sand actually contained in 1 m³ of this damp sand is:

(a) 0.80 m³ (b) 0.75 m³ (c) 1.25 m³ (d) 0.25 m³

32. The quantity of cement required for 0.5 m³ of 1 : 3 cement mortar (dry volume = 1.3 × wet volume; density of cement 1440 kg/m³) is about:

(a) 163 kg (b) 180 kg (c) 234 kg (d) 312 kg

33. S1: Using a finer sand increases the water demand of a mortar.

S2: Sea sand is preferred for plastering because the salts in it retain moisture.

(a) Both S1 and S2 are true (b) S1 is true and S2 is false (c) S1 is false and S2 is true (d) Both S1 and S2 are false

34. Match List-I with List-II:

List-I	List-II
A. Quick lime	1. $Ca(OH)_2$
B. Slaked lime	2. Contains clay and sets under water
C. Fat lime	3. CaO
D. Hydraulic lime	4. Nearly pure lime that sets by carbonation in air

(a) A-3, B-1, C-4, D-2 (b) A-1, B-3, C-4, D-2 (c) A-3, B-1, C-2, D-4 (d) A-4, B-1, C-3, D-2

35. Conical holes that appear in a plastered surface, caused by particles in the mix that expand after the plaster has set, are called:

(a) blistering (b) crazing (c) flaking (d) popping

H. Concrete: Ingredients, Admixtures and Water–Cement Ratio

36. A mix is batched per 50 kg bag of cement at a water–cement ratio of 0.45. If the damp aggregates already carry 2.5 litres of surface water, the water to be added at the mixer is:

(a) 22.5 litres (b) 25.0 litres (c) 20.0 litres (d) 18.0 litres

37. Assertion (A): For the same workability, rounded aggregates need less cement paste than angular aggregates.

Reason (R): For a given volume, rounded particles have the least surface area and pack with fewer voids.

(a) Both A and R are true and R is the correct explanation of A (b) Both A and R are true, but R is not the correct explanation of A

(c) A is true, but R is false (d) Both A and R are false

38. Study the statement and the conclusions and select the most appropriate option.
Statement: Calcium chloride is added as an admixture to the concrete of a reinforced concrete beam.
Conclusion 1: The concrete gains early strength faster.
Conclusion 2: The risk of corrosion of the reinforcement increases.
(a) Only conclusion 1 is correct (b) Only conclusion 2 is correct
(c) Both conclusions 1 and 2 are correct (d) Neither conclusion 1 nor 2 is correct

39. As per IS 456 : 2000, for reinforced concrete under 'moderate' exposure, the maximum free water–cement ratio and the minimum grade of concrete are:
(a) 0.55 and M20 (b) 0.50 and M25 (c) 0.45 and M30 (d) 0.40 and M40

40. Which of the following statements is INCORRECT?
(a) Coarse aggregate is retained on the 4.75 mm sieve (b) Zone IV sand is finer than Zone I sand
(c) A higher fineness modulus indicates a finer aggregate (d) A well-graded aggregate has the minimum voids

I. Fresh and Hardened Concrete; Non-destructive Testing

41. A concrete cylinder 150 mm in diameter and 300 mm long fails in the split tensile test at a load of 45π kN (about 141 kN). The split tensile strength is:
(a) 1.0 N/mm² (b) 2.0 N/mm² (c) 3.14 N/mm² (d) 6.28 N/mm²

42. As per IS 456 : 2000, the short-term static modulus of elasticity of concrete with a characteristic strength of 36 N/mm² is:
(a) 30,000 N/mm² (b) 25,000 N/mm² (c) 36,000 N/mm² (d) 18,000 N/mm²

43. Match List-I (test) with List-II (purpose):

List-I	List-II
A. Half-cell potential test	1. Depth of carbonation
B. Cover meter survey	2. Surface hardness of concrete
C. Phenolphthalein spray	3. Likelihood of corrosion of reinforcement
D. Rebound hammer test	4. Location and cover of reinforcing bars

(a) A-4, B-3, C-1, D-2 (b) A-3, B-4, C-2, D-1 (c) A-1, B-4, C-3, D-2 (d) A-3, B-4, C-1, D-2

44. In an ultrasonic pulse velocity test by direct transmission, the pulse takes 75 microseconds to travel through a 300 mm thick concrete wall. The pulse velocity and the quality grading as per IS 13311 are:
(a) 2.5 km/s, doubtful (b) 4.0 km/s, excellent (c) 0.25 km/s, doubtful (d) 4.0 km/s, good

45. S1: The compacting factor test is more suitable than the slump test for concrete of low workability.
S2: For the compression test, a 150 mm cube mould is filled in four layers, each tamped with 25 strokes.
(a) Both S1 and S2 are true (b) S1 is true and S2 is false (c) S1 is false and S2 is true (d) Both S1 and S2 are false

J. Mix Design and Steel

46. The target mean strength for M20 concrete, with the standard deviation taken as 4.0 N/mm², is:
(a) 24.0 N/mm² (b) 21.65 N/mm² (c) 26.6 N/mm² (d) 28.25 N/mm²

47. For 20 mm aggregate the water content for a 50 mm slump is 186 kg/m³, to be increased by 3 % for every additional 25 mm of slump. For a slump of 100 mm and a water–cement ratio of 0.50, the cement content is about:
(a) 372 kg/m³ (b) 383 kg/m³ (c) 450 kg/m³ (d) 394 kg/m³

48. Assertion (A): Better quality control at site allows a lower target mean strength for the same grade of concrete.
Reason (R): The target mean strength is the characteristic strength plus 1.65 times the standard deviation, and good control reduces the standard deviation.
(a) Both A and R are true and R is the correct explanation of A (b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false (d) Both A and R are false

49. Match List-I (material) with List-II (approximate carbon content):

List-I	**List-II**
A. Wrought iron	1. 2 to 4 %
B. Mild steel	2. 0.60 to 1.5 %
C. High carbon steel	3. Less than 0.15 %
D. Cast iron	4. 0.15 to 0.30 %

(a) A-3, B-4, C-2, D-1 (b) A-4, B-3, C-2, D-1 (c) A-3, B-4, C-1, D-2 (d) A-1, B-4, C-2, D-3

50. Study the statement and the conclusions and select the most appropriate option.
Statement: The carbon content of a steel is raised from 0.2 % to 0.6 %.
Conclusion 1: Its weldability improves.
Conclusion 2: Its modulus of elasticity increases considerably.
(a) Only conclusion 1 is correct (b) Only conclusion 2 is correct
(c) Both conclusions 1 and 2 are correct (d) Neither conclusion 1 nor 2 is correct

Answer key with short solutions

A. Timber: Structure, Moisture and Strength

1. (a) Both are true and R explains A. Shrinkage is greatest tangentially, about half of that radially and almost nil longitudinally.

2. (b) Moisture content is on the oven-dry weight: wet = dry × 1.20, so dry = 1.80/1.20 = 1.50 kg. (1.44 kg wrongly takes 20 % of the wet weight.)

3. (c) Cambium → growing layer under the bark; pith → centre; medullary rays → radial bands; sapwood → outer, light, sap-carrying rings.

4. (a) Below the fibre saturation point the cell walls lose bound water: strength rises and the timber shrinks. So conclusion 2 is wrong.

5. (d) Group A: E above 12.6 kN/mm²; Group B: 9.8 to 12.6 kN/mm²; Group C: 5.6 to 9.8 kN/mm².

B. Timber: Defects, Seasoning, Preservation and Plywood

6. (c) S1 is false: the dry-rot fungus needs dampness and lack of ventilation; 'dry' refers to the powder it leaves. S2 is true.

7. (d) Bow → curved along the length; cup → curved across the width; honeycombing → internal cracks; case hardening → hard dry surface over a wet core.

8. (b) ASCU = arsenic pentoxide : copper sulphate : potassium dichromate = 1 : 3 : 4 by weight. It is odourless, water-borne and can be painted over.

9. (d) The grain of adjacent veneers is at right angles (cross-banded). That is what gives plywood its equal strength in both directions and its dimensional stability.

10. (c) A is true: air-dried timber only comes to equilibrium with the surrounding air. R is false: it describes kiln seasoning; air seasoning is done in open stacks under a shed.

C. Bricks

11. (b) Volume = 4 × 3 × 0.2 = 2.4 m³. A modular brick with mortar is 200 × 100 × 100 mm, so 500 bricks per m³. 2.4 × 500 = 1200.

12. (a) Chuffs → rain on hot bricks; efflorescence → alkali salts; black core → unburnt carbon; lime blowing → lime nodules that slake and swell.

13. (a) Both are true. Alumina gives plasticity but too much causes shrinkage and warping; silica prevents shrinkage but too much destroys cohesion and makes bricks brittle.

14. (d) Cold absorption = 0.35 kg; boiling absorption = 0.45 kg. Saturation coefficient = 0.35/0.45 = 0.78. (0.97 wrongly divides the wet weights.)

15. (a) The class number is the average compressive strength in N/mm². The absorption limit is 20 % up to class 12.5 and 15 % only for higher classes, so conclusion 2 is wrong.

D. Brick Masonry and Building Stones

16. (c) A is true: very strong mortar is costly and brittle and gives little extra strength. R is false: masonry strength rises only slightly with mortar strength; brick strength governs.

17. (b) Slenderness ratio = effective height ÷ effective thickness = 3000/200 = 15. IS 1905 limits it to 27 for load-bearing walls in cement mortar.

18. (d) Granite → igneous, siliceous; sandstone → sedimentary, siliceous; slate → metamorphic from shale (argillaceous); marble → metamorphic from limestone (calcareous).

19. (b) Sodium sulphate crystallises in the pores as the stone dries and imitates the expansion of freezing water. The loss of weight measures frost resistance.

20. (d) Headers and stretchers alternating in the same course is Flemish bond. English bond has alternate courses of headers and stretchers.

E. Cement: Compounds, Hydration and Types

21. (a) C_3S hydrates fast and gives early strength, with about twice the heat of C_2S. So P is faster but hotter; conclusion 2 is wrong.

22. (c) Rapid hardening → early strength for repairs; low heat → mass concrete; sulphate resisting (C_3A below 5 %) → sulphate soils; quick setting → under or in running water.

23. (c) C_3A hydrates almost at once, giving about 865 J/g. Gypsum is added to control it; without gypsum it causes flash set.

24. (b) S1 is true. S2 is false: the pozzolanic reaction is slow, so early strength is lower and longer curing is needed; later strength and durability are better.

25. (a) Both are true and R explains A. Converting the leachable calcium hydroxide into C–S–H gel is what improves the durability of PPC concrete.

F. Cement: Tests

26. (c) Water = (P/4 + 3) % of the combined weight of cement and sand = (7 + 3) % of 800 g = 80 g. (56 g is 28 % of the cement alone.)

27. (d) Initial set ≥ 30 minutes; final set ≤ 600 minutes; Le Chatelier expansion ≤ 10 mm; autoclave expansion ≤ 0.8 %.

28. (c) 43 grade: 23, 33 and 43 N/mm² at 3, 7 and 28 days. (27 and 37 are the 3- and 7-day values of 53 grade.)

29. (b) 10 bags = 500 kg. Volume = 500/1440 = 0.347 m³, that is about 35 litres per bag.

30. (b) The Le Chatelier test measures unsoundness due to free lime; 14 mm exceeds the 10 mm limit. It says nothing about fineness, which is tested by sieving or by Blaine's apparatus.

G. Mortar and Lime

31. (a) Bulked volume = 1.25 × dry volume, so dry volume = 1/1.25 = 0.80 m³. (0.75 m³ wrongly subtracts 25 % of the bulked volume.)

32. (c) Dry volume = 0.65 m³. Cement = 0.65/4 = 0.1625 m³ = 0.1625 × 1440 = 234 kg. (180 kg omits the factor 1.3; 312 kg divides by 3 instead of 4 parts.)

33. (b) S1 is true: finer sand has more surface to wet. S2 is false: salts in sea sand attract moisture and cause dampness and efflorescence, so sea sand is avoided.

34. (a) Quick lime is CaO; slaking with water gives $Ca(OH)_2$; fat lime is high-calcium lime hardening by carbonation; hydraulic lime contains clay and sets under water.

35. (d) Popping: a particle such as unslaked lime expands and pushes out a small cone of plaster. Blistering is a local swelling; crazing is a network of hair cracks.

H. Concrete: Ingredients, Admixtures and Water–Cement Ratio

36. (c) Total water = 0.45 × 50 = 22.5 litres. Water already in the aggregates counts, so add 22.5 − 2.5 = 20 litres.

37. (a) Both are true and R explains A. Angular aggregates need more paste but give better bond, which is why they are used for high-strength concrete.

38. (c) Calcium chloride is an accelerator, but chloride ions break down the protective film on steel. That is why it is not permitted in reinforced and prestressed concrete.

39. (b) Mild: 0.55, M20. Moderate: 0.50, M25. Severe: 0.45, M30. Very severe: 0.45, M35. Extreme: 0.40, M40.

40. (c) Fineness modulus is the sum of cumulative percentages retained ÷ 100, so a higher value means more material retained on the sieves, i.e. a coarser aggregate.

I. Fresh and Hardened Concrete; Non-destructive Testing

41. (b) $f = 2P/(\pi DL) = 2 \times 45{,}000\pi/(\pi \times 150 \times 300) = 90{,}000/45{,}000 = 2.0$ N/mm².

42. (a) $E = 5000\sqrt{fck} = 5000 \times 6 = 30{,}000$ N/mm².

43. (d) Half-cell potential → corrosion activity; cover meter → bar position and cover; phenolphthalein → carbonation depth (pink where still alkaline); rebound hammer → surface hardness.

44. (d) Velocity = 0.300 m ÷ 75 × 10^{-6} s = 4000 m/s = 4.0 km/s. Grading: above 4.5 excellent; 3.5–4.5 good; 3.0–3.5 medium; below 3.0 doubtful.

45. (b) S1 is true. S2 is false: the cube is filled in three layers of 35 strokes; four layers of 25 strokes is the rule for the slump cone.

J. Mix Design and Steel

46. (c) $f'ck = fck + 1.65\,s = 20 + 1.65 \times 4 = 26.6$ N/mm². (24.0 adds s only once; 28.25 uses s = 5.)

47. (d) Two steps of 25 mm → 6 % more water: 186 × 1.06 = 197 kg/m³. Cement = 197/0.50 = 394 kg/m³. (372 ignores the slump correction.)

48. (a) Both are true and R explains A. A smaller margin means less cement for the same grade.

49. (a) Carbon rises in the order wrought iron (below 0.15 %), mild steel (0.15–0.30 %), high carbon steel (0.60–1.5 %), cast iron (2–4 %).

50. (d) More carbon raises strength and hardness but lowers ductility and weldability. The modulus of elasticity stays at about 2×10^5 N/mm² for all steels. Both conclusions are wrong.

Score analysis — fill this in and revise the weakest section first

Section	Questions	Max	Correct	Wrong	Net (C − W/3)
A. Timber: Structure, Moisture and Strength	1–5	5			
B. Timber: Defects, Seasoning, Preservation and Plywood	6–10	5			
C. Bricks	11–15	5			
D. Brick Masonry and Building Stones	16–20	5			
E. Cement: Compounds, Hydration and Types	21–25	5			
F. Cement: Tests	26–30	5			
G. Mortar and Lime	31–35	5			
H. Concrete: Ingredients, Admixtures and Water–Cement Ratio	36–40	5			
I. Fresh and Hardened Concrete; Non-destructive Testing	41–45	5			
J. Mix Design and Steel	46–50	5			
Total	1–50	**50**			

Net 35 or more: ready, move to the next module. **25–35:** re-read the weak topics and redo their quizzes. **Below 25:** 3–4 more days on Module 4, then retake after a week.